Power entry, limiter and isolated converter

File: power_entry.kicad_sch

5 V rails, switches and supervision

File: rails.kicad_sch

Compute module CM5

File: cm5.kicad_sch

Host interfaces: CAN and RS-422

File: interfaces.kicad_sch

Housekeeping ADC, reference, rail sensing and thermistors

File: housekeeping.kicad_sch

Maintenance port and head link

File: maint_head.kicad_sch

POWER DOMAINS
BUS_IN / BUS_F / BUS_SW / BUS_FILT, PRI_RTN: spacecraft bus side of the CCG10 isolation barrier.
5V_BUS: CCG10 output, always on while the payload is powered.
3V7_A: always-on 3.71 V (shunt amplifiers, watchdog, enable pull-ups). VREF: 4.096 V reference (ADC VA, thermistor bias).
5V_CM5 (SW1) and 5V_HEAD (SW2): switched rails. CM5_3V3: the CM5's own 3.3 V output, on only while the CM5 runs.
Every part that drives a CM5 pin is powered from CM5_3V3 or 5V_RS422 (gated by CM5_3V3), so no CM5 pin is driven before its 5 V is up (CM5 datasheet, section 3).
GND: payload ground (CCG10 secondary). CHASSIS: rods and housing; tied to GND only through the DNP R14/C19.
Commercial parts are fitted where the space-grade variant is not buyable (MPN_ALT names it). Derating: NASA EEE-INST-002 (README).

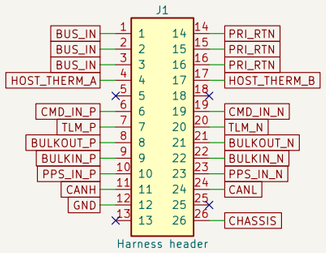
Pin tables from vendor datasheets; sources in the design record.
SLANT carrier board (Raspberry Pi CM5).
Carthage College, SLANT

Sheet: /
File: slant_carrier.kicad_sch

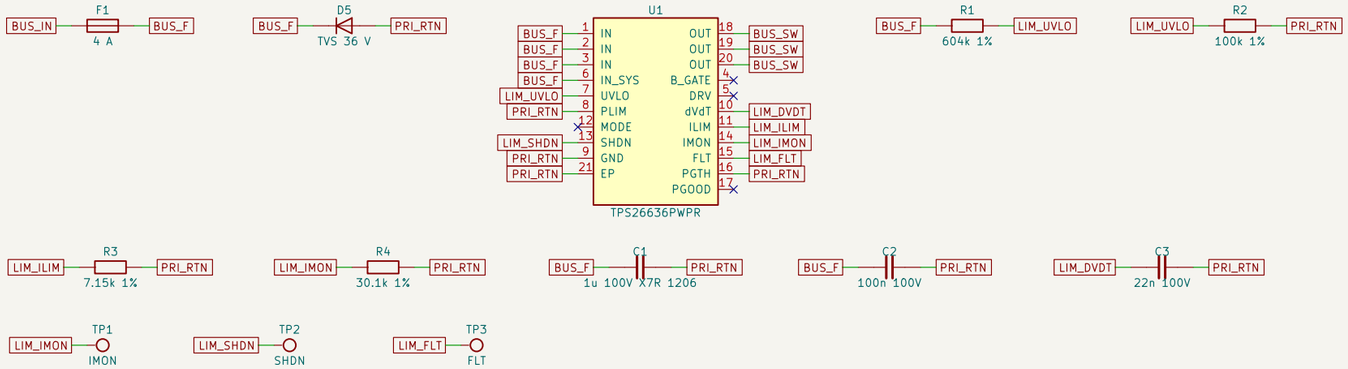
Title: SLANT carrier board (CM5): sheet index

Size: A3	Date: 2026-10-10	**Rev: A4-5**
KiCad E.D.A. 10.0.7		Id: 1/7

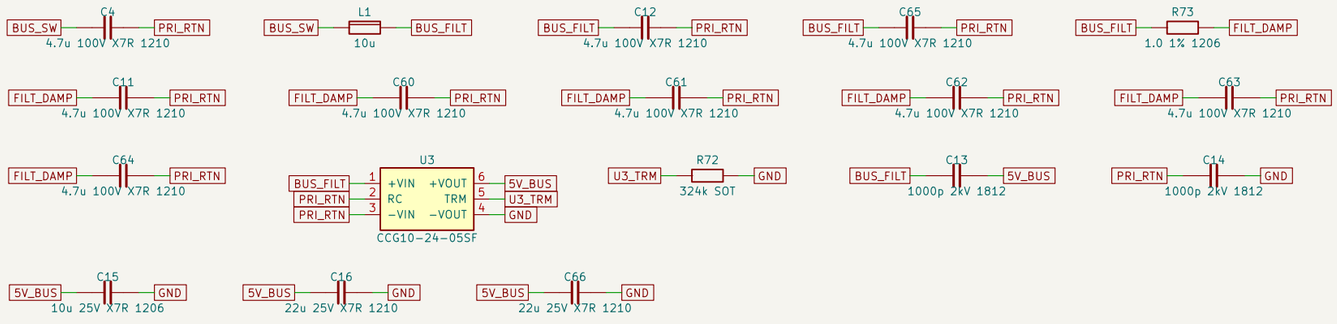
Harness header (Harwin Gecko 26-way, host power and data)



Fuse, TVS and latching current limiter (bus side)



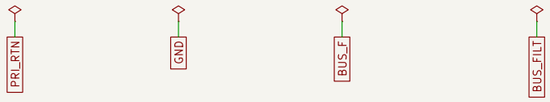
Input filter (TDK CCG10 Class A, damped ceramic) and isolated converter



Chassis and mounting holes



Power flags for ERC



NOTES (value basis and items to confirm)

J1: Gecko 1.25 mm, 2 x 13, through-hole. Three contacts per power rail: 1.25 A max input / 3 = 0.42 A per contact (2 A per contact simultaneous rating; EEE-INST-002 derates to the wire gauge, 26 AWG 1.4 A). Harwin numbers row by row (specification C125XX issue 15, A1.2): 1-13 on one row, 14-26 on the other, 14 across from 1; the 26-way order is extrapolated from Harwin's 10-way drawing. Each pair sits across the rows (6/19 ... 11/24); bus and return across from each other (1-3/14-16). Isolation domains: host side in columns 1-4 (bus, return, host thermistor), column 5 empty, payload in columns 6-12, chassis at pin 26; 1.6 mm pad gap between host and payload pins, 0.87 mm between payload GND and chassis (500 Vdc after conformal coat, IPC-2221B A7 0.8 mm). Latch barbs are specified for 1.6 mm boards (drawing G125-MVXXX05LXP issue 8); the board is 2.4 mm: confirm retention with Harwin or solder the barbs.

F1: TDK requires an external fuse of 4 A or lower for CCG10-24 (instruction manual C286-04-11E). EEE-INST-002: <= 50 % of rating, less 0.2 %/C above 25 C: at 60 C, 1.72 A allowed against 1.25 A maximum input.

D5: After the fuse, so a shorted TVS opens F1. Clamp stays under the TPS26636's 67 V absolute maximum. Rating TBC against the host surge specification.

U1: MODE open: latch-off after an overload; the host's per-pass power cycle (or SHDN low) resets it. IN_SYS tied to IN (no blocking FET); B_GATE, DRV floating. PLIM to GND: power limit off. PGTH to GND: PGOOD unused. Output clamp 35.7-39 V protects the CCG10 from bus surges. Recommended input 4.5-60 V: 36 V bus is 60 %.

R1: UVLO divider: rising 1.2 V x (604k + 100k)/100k = 8.45 V (8.28-8.62 V); falling 1.122 V x 7.04 = 7.90 V. Above 300k as the datasheet requires. 51 uA at 36 V.

R3: I_OL = 18 / R_ILIM[k] = 2.52 A (range 3-30k). Maximum payload input 1.25 A at 9 V (10 W).

R4: IMON 27.9 uA/A x 30.1k = 0.84 V/A; 2.1 V at the limit, under the 4 V maximum. No capacitor on IMON (datasheet). Bus side: test point only.

C1: CIN >= 1 uF (TPS2663 surge guidance)

C2: IN_SYS >= 0.1 uF

C3: t_dVdT = 20.8e3 x VIN x C = 12.8 ms at 28 V; inrush into about 52 uF of filter capacitance 0.11 A

TP2: SHDN internally pulled up (2.7 V open circuit). Pull low and release to reset a latched trip on the bench. Keep the trace short.

TP3: Open drain, active low; bench pull-up only (no bus-side logic supply)

C4: TDK quick manual Sec 8 filter C1 position, now ceramic; also the TPS2663 output capacitor (>= 0.1 uF)

L1: TDK quick manual Sec 8 filter L1, 10 uH

C12: TDK filter C2 (10 uF) as 2 x 4.7 uF 100 V: 36 V is 36 % of rated

R73: Damping leg across C2, replacing the electrolytic's ESR. Z0 = sqrt(10 uH / 9.4 uF) = 1.03 Ohm; with Cd/C = 3, the Middlebrook optimum R is about 0.75 Ohm. 1.0 Ohm chosen; filter output impedance to confirm against the CCG10 input impedance on the bench.

C11: Damping capacitor bank, 6 x 4.7 uF = 28 uF (ceramic, because wet aluminum is not listed in EEE-INST-002)

U3: RC tied to -Vin: always on (negative logic). Trimmed up through TRM (R72). Pin positions from quick manual Sec 8 top view.

R72: Rb, TRM to -Vout, trim up: Vout = 5.02 V + 10.10 / (15 + Rb[k]) (TDK manual Table 6-2) = 5.05 V at 324k. Select on test per converter for 5.05-5.08 V measured: the +-2 % setpoint tolerance is larger than the trim. Stay below 5.25 V, TDK's 105 % OVP minimum.

C13: TDK quick manual Sec 8 filter C3, +Vin to +Vout across the barrier

C14: TDK quick manual Sec 8 filter C4, -Vin to -Vout across the barrier

C15: TDK quick manual Sec 8 filter C5

C16: 5 V bulk, 2 x 22 uF ceramic (was 100 uF tantalum); output capacitance stays under TDK's 470 uF maximum

R14: Payload ground to chassis, not fitted: the host sets the single-point ground

Pin tables from vendor datasheets; sources in the design record.
SLANT carrier board (Raspberry Pi CM5).

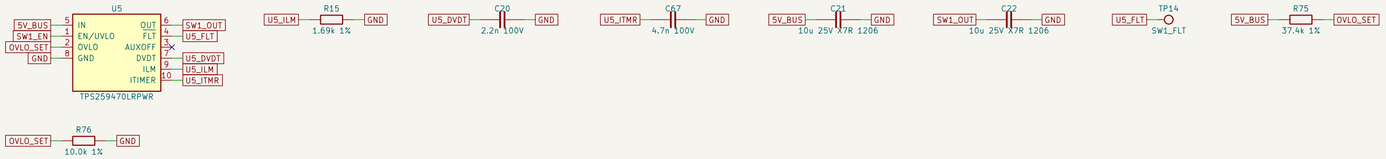
Carthage College, SLANT

Sheet: /Power entry, limiter and isolated converter/
File: power_entry.kicad_sch

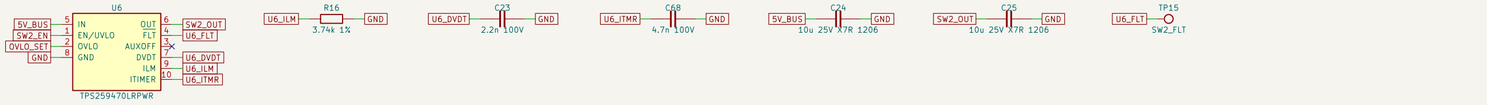
Title: Power entry, limiter and isolated converter

Size: A3	Date: 2026-10-10	**Rev: A4-5**
KiCad E.D.A. 10.0.7		Id: 2/7

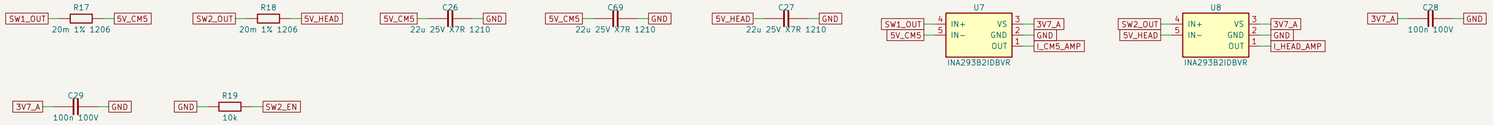
SW1: CM5 5 V rail switch (and shared OVLO divider)



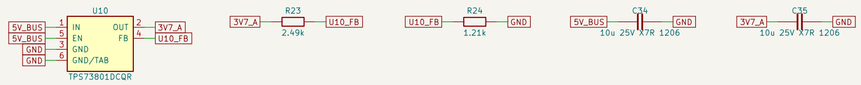
SW2: head 5 V rail switch



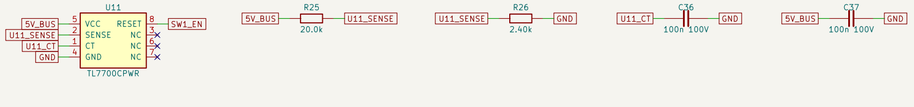
Rail shunts and current amplifiers



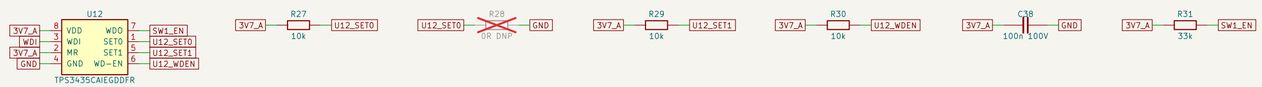
Always-on 3.71 V (amplifiers, watchdog, enable pull-ups)



Supervisor: holds SW1 off until 5 V is good



Watchdog: opens SW1 when not serviced



Test points



Power flags for ERC

NOTES (value basis and items to confirm)
U5: SW1, CM5 rail. L variant: latches off after an overcurrent fault until EN/UVLO is cycled (the supervisor and watchdog do this) or power is cycled. Reverse blocking when off.
AUXOFF unused (open drain, floating).
R15: R_ILM = 3334 / I_LIM: 1.97 A (1.78–2.17 A, +–10 %). Above the converter's 2 A at the high end: a CM5 fault may hiccup the CCG10 first, which also power-cycles the CM5. Set from the bench boot-peak current. ILM also carries the current monitor, 182 uA/A.
C20: SR = 2000 / C[pF] = 0.91 V/ms: 5 V in 5.5 ms; inrush 0.91 V/ms x C_out
C67: Overcurrent blanking t = 1.51 V x 4.7 nF / 1.8 uA = 3.9 ms, then latch-off
C21: SW1 IN
C22: SW1 OUT
TP14: Open drain, active low: bench pull-up
R75: OVLO divider shared by SW1 and SW2: rising 1.20 V x 4.74 = 5.69 V (5.61–5.80 V); falling 1.09 V x 4.74 = 5.17 V. Opens both switches if the converter output runs high.
U6: SW2, head rail. Same configuration as SW1.
R16: R_ILM: 0.89 A (about 0.80–0.98 A; TI states +–10 % above 1 A only). Head 3.4 W uncooled = 0.68 A at 5 V (xIJ manual). Max 0.98 A + CM5 0.9 A stays under the CCG10's 2 A, so a head fault trips SW2 before the converter hiccups.
R17: CM5 rail shunt: 20 mV/A. 2.17 A limit maximum: 94 mW, 38 % of 0.25 W (metal strip limit 50 %).
R18: Head rail shunt
C26: CM5 bulk, 2 x 22 uF ceramic (was 47 uF tantalum)
C27: Head bulk, ceramic (was 22 uF tantalum)
U7: B pinout. 50 V/V x 20 mOhm = 1.0 V/A: output swings to VS – 0.15 V = 3.56 V (3.56 A)
C28: U7 VS
C29: U8 VS
R19: Head off until the CM5 drives SW2_EN high (GPIO16). Against an RP1 boot pull-up of about 50k, EN slts at 0.55 V, under the 1.08 V falling threshold minimum.
U10: Always-on 3.71 V: shunt amplifiers, watchdog, SW1_EN pull-up and watchdog straps (the 3.3 V module TPSM82823 has no stock until 4/2027 and is removed). Above CM5_3V3 max (3.465 V).
R23: R2 (top): VOUT = 1.21 V x (1 + 2.49k/1.21k) + 3 uA x 2.49k = 3.71 V (derived)
R24: R1 (bottom) < 4.17k
C35: >= 10 uF, ESR <= 3 Ohm
U11: Holds SW1 off until 5V_BUS is above ~4.67 V, then 10 ms more
R25: R1: hysteresis = Is x 20.0k = 0.03–0.07 V (Is 1.5–3.5 uA, –40 to 85 C)
R26: R2: falling trip = Vs x 22.4/2.40: 4.67 V typ, 4.57–4.76 V (Vs 490–510 mV, –40 to 85 C); release at most 4.83 V, below the trimmed 5V_BUS minimum of 5.02 V. The trim gives the CM5 its margin: the supervisor catches brown-outs.
C36: tpo = CT x 1e5 s = 10 ms
U12: Open-drain, no startup delay. 1 s base x 64 (SET1:SET0 = 11) = 64 s, min 57.6 s. WDO pulse 200 ms. VDD 3.71 V (1.04–6 V): WDI VIH = 0.7 x VDD = 2.60 V, met by a 3.3 V CM5 GPIO. Only WDI connects to the CM5.
R27: SET0 high
R28: Fit and remove R27 for 32 s
R29: SET1 high
R30: Watchdog always enabled
R31: EN pull-up for the wired TL7700 RESET and TPS3435 WDO (both open drain). 0.11 mA sink; EN/UVLO falling threshold 1.08 V minimum, abs max 6.5 V.

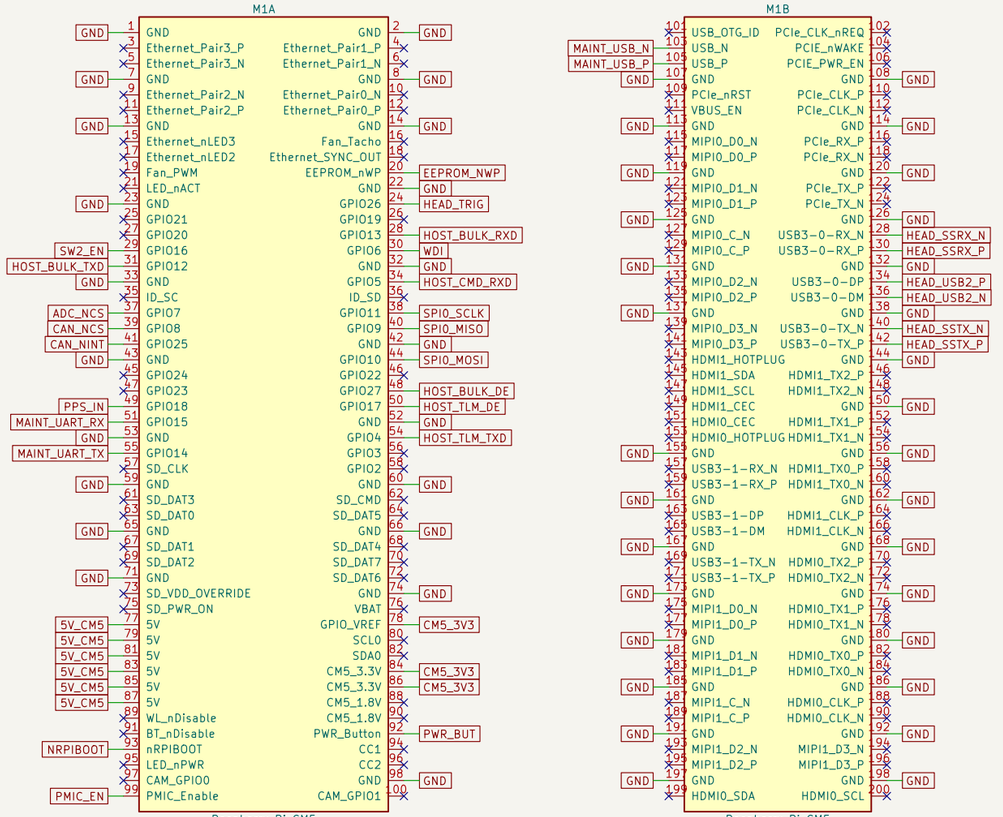
Pin tables from vendor datasheets; sources in the design record.
SLANT carrier board (Raspberry Pi CM5).
Carthage College, SLANT

Sheet: /5 V rails, switches and supervision/
File: rails.kicad_sch

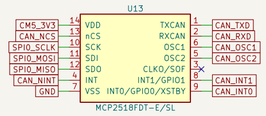
Title: 5 V rails, switches and supervision

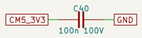
Size: A2	Date: 2026-10-10	**Rev: A4-5**
KiCad E.D.A. 10.0.7		Id: 3/7

Raspberry Pi Compute Module 5 (both connectors)


M1A
GND 1
Ethernet_Pair3_P 3
Ethernet_Pair3_N 5
GND 7
Ethernet_Pair2_N 9
Ethernet_Pair2_P 11
GND 13
Ethernet_nLED3 15
Ethernet_nLED2 17
Fan_PWM 19
LED_nACT 21
GND 23
GPIO21 25
GPIO20 27
GPIO16 29
GPIO12 31
GND 33
ID_SC 35
GPIO7 37
GPIO8 39
GPIO25 41
GND 43
GPIO24 45
GPIO23 47
GPIO18 49
GPIO15 51
GND 53
GPIO14 55
SD_CLK 57
GND 59
SD_DAT3 61
SD_DAT0 63
GND 65
SD_DAT1 67
SD_DAT2 69
GND 71
SD_VDD_OVERRIDE 73
SD_PWR_ON 75
5V 77
5V 79
5V 81
5V 83
5V 85
5V 87
WL_nDisable 89
BT_nDisable 91
nRPIBOOT 93
LED_nPWR 95
CAM_GPIO0 97
PMIC_Enable 99
GND 2
Ethernet_Pair1_P 4
Ethernet_Pair1_N 6
GND 8
Ethernet_Pair0_N 10
Ethernet_Pair0_P 12
GND 14
Fan_Tacho 16
Ethernet_SYNC_OUT 18
EEPROM_nWP 20
GND 22
GPIO26 24
GPIO19 26
GPIO13 28
GPIO6 30
GND 32
GPIO5 34
ID_SD 36
GPIO11 38
GPIO9 40
GND 42
GPIO10 44
GPIO22 46
GPIO27 48
GPIO17 50
GND 52
GPIO4 54
GPIO3 56
GND 58
SD_CMD 60
SD_DAT5 62
GND 64
SD_DAT4 66
SD_DAT7 68
SD_DAT6 70
GND 72
VBAT 74
GPIO_VREF 76
SCL0 78
SDA0 80
CM5_3.3V 82
CM5_3.3V 84
CM5_1.8V 86
CM5_1.8V 88
PWR_Button 90
CC1 92
CC2 94
GND 96
CAM_GPIO1 98
GND 100
GND
GND
GND
EEPROM_NWP
GND
HEAD_TRIG
HOST_BULK_RXD
WDI
GND
HOST_CMD_RXD
SPI0_SCLK
SPI0_MISO
GND
SPI0_MOSI
HOST_BULK_DE
HOST_TLM_DE
GND
HOST_TLM_TXD
GND
GND
GND
GND
CM5_3V3
CM5_3V3
CM5_3V3
PWR_BUT
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SW2_EN
HOST_BULK_TXD
GND
ADC_NCS
CAN_NCS
CAN_NINT
GND
PPS_IN
MAINT_UART_RX
GND
MAINT_UART_TX
GND
PMIC_EN
Raspberry Pi CM5
M1B
USB_OTG_ID 101
USB_N 103
USB_P 105
GND 107
PCIe_nRST 109
VBUS_EN 111
GND 113
MIPI0_D0_N 115
MIPI0_D0_P 117
GND 119
MIPI0_D1_N 121
MIPI0_D1_P 123
GND 125
MIPI0_C_N 127
MIPI0_C_P 129
GND 131
MIPI0_D2_N 133
MIPI0_D2_P 135
GND 137
MIPI0_D3_N 139
MIPI0_D3_P 141
HDMI1_HOTPLUG 143
HDMI1_SDA 145
HDMI1_SCL 147
HDMI1_CEC 149
HDMI0_CEC 151
HDMI0_HOTPLUG 153
GND 155
USB3-1-RX_N 157
USB3-1-RX_P 159
GND 161
USB3-1-DP 163
USB3-1-DM 165
GND 167
USB3-1-TX_N 169
USB3-1-TX_P 171
GND 173
MIPI1_D0_N 175
MIPI1_D0_P 177
GND 179
MIPI1_D1_N 181
MIPI1_D1_P 183
GND 185
MIPI1_C_N 187
MIPI1_C_P 189
GND 191
MIPI1_D2_N 193
MIPI1_D2_P 195
GND 197
HDMI0_SDA 199
PCIe_CLK_nREQ 102
PCIE_nWAKE 104
PCIE_PWR_EN 106
GND 108
PCIe_CLK_P 110
PCIe_CLK_N 112
GND 114
PCIe_RX_P 116
PCIe_RX_N 118
GND 120
PCIe_TX_P 122
PCIe_TX_N 124
GND 126
USB3-0-RX_N 128
USB3-0-RX_P 130
GND 132
USB3-0-DP 134
USB3-0-DM 136
GND 138
USB3-0-TX_N 140
USB3-0-TX_P 142
GND 144
HDMI1_TX2_P 146
HDMI1_TX2_N 148
GND 150
HDMI1_TX1_P 152
HDMI1_TX1_N 154
GND 156
HDMI1_TX0_P 158
HDMI1_TX0_N 160
GND 162
HDMI1_CLK_P 164
HDMI1_CLK_N 166
GND 168
HDMI0_TX2_P 170
HDMI0_TX2_N 172
GND 174
HDMI0_TX1_P 176
HDMI0_TX1_N 178
GND 180
HDMI0_TX0_P 182
HDMI0_TX0_N 184
GND 186
HDMI0_CLK_P 188
HDMI0_CLK_N 190
GND 192
MIPI1_D3_N 194
MIPI1_D3_P 196
GND 198
HDMI0_SCL 200
MAINT_USB_N
MAINT_USB_P
GND
GND
GND
GND
GND
GND
GND
GND
GND
GND
HEAD_SSRX_N
HEAD_SSRX_P
GND
HEAD_USB2_P
HEAD_USB2_N
GND
HEAD_SSTX_N
HEAD_SSTX_P
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GND
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GND

CAN FD controller (SPI)



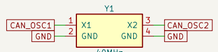


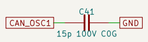


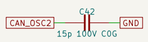






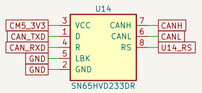






NOTES (value basis and items to confirm)
U13: Pinout from Microchip DS20006027B, read from a SparkFun-hosted copy because Microchip's site refuses automated access
R32: INT configured open drain (IOCON.INTOD)
R33: Deselected while the CM5 boots
R70: INT1/GPIO1 reset as a GPIO input: pulled up rather than left floating
R71: INT0/GPIO0/XSTBY reset as a GPIO input: pulled up
Y1: 40 MHz, CL 12 pF, +-20 ppm, +-50 ppm over -40 to 125 C, ESR 35 Ohm (Abracon page). Pad assignment (2 and 4 ground) to confirm on the ABM8AIG drawing.
C41: C = 2 x (CL - Cstray) = 2 x (12 - 4.5) = 15 pF; trim on the bench for frequency
C44: Per SEP EVM decoupling
R34: Slope control, about 15 V/us (10k-100k range)
D1: Place at the harness header
R35: Split termination, fit only if the payload ends the bus
U15: Host link: driver 1 telemetry out, driver 2 bulk out; receiver 1 command in, receiver 2 bulk in. Thin distributor stock: buy early.
U18: Gates the RS-422 5 V with the CM5's own 3.3 V: the receivers cannot drive CM5 GPIOs before CM5_3V3 is up. QOD tied to VOUT: internal 24 Ohm discharge when CM5_3V3 falls. ON VIH 1 V: smart pull-down. Pin 4 NC, left floating (datasheet).
C58: U18 CIN 1 uF
C59: U18 COUT: rise about 1.3 ms at 5 V, inrush 10 uF x 5 V / 1.3 ms = 39 mA
R37: Driver off until the CM5 enables it
R39: Idle mark
D4: Feeds the fail-safe bias from 5V_RS422 and blocks reverse current: a host driving the lines while the CM5 is off cannot back-feed 5V_RS422 and the CM5's 5 V pins. 6.1 mA load against 70 mA IF (50 % derating allows 35 mA): 5 V against 70 V VR.
R43: Receiver termination (cable impedance TBC), 1206 (0.25 W): 3.5 V differential gives 0.10 W, 41 %.
R41: Fail-safe bias: (5 V - 0.7 V diode) x 120/2120 = 0.24 V across the termination (> 0.2 V)
R44: 5 V receiver output to 3.3 V GPIO, ratio 0.6: 5.5 V gives 3.3 V; 3.8 V VOH min gives 2.28 V > 2.0 V VIH. Command and PPS links 10k/15k (low rates).
R48: Receiver termination (cable impedance TBC), 1206 (0.25 W): 3.5 V differential gives 0.10 W, 41 %.
R46: Fail-safe bias: (5 V - 0.7 V diode) x 120/2120 = 0.24 V across the termination (> 0.2 V)
R49: 5 V receiver output to 3.3 V GPIO, ratio 0.6: 5.5 V gives 3.3 V; 3.8 V VOH min gives 2.28 V > 2.0 V VIH. Bulk link 2.2k/3.3k for edge rate.
R53: Receiver termination (cable impedance TBC), 1206 (0.25 W): 3.5 V differential gives 0.10 W, 41 %.
R51: Fail-safe bias: (5 V - 0.7 V diode) x 120/2120 = 0.24 V across the termination (> 0.2 V)
R54: 5 V receiver output to 3.3 V GPIO, ratio 0.6: 5.5 V gives 3.3 V; 3.8 V VOH min gives 2.28 V > 2.0 V VIH. Command and PPS links 10k/15k (low rates).
U16: PPS receiver; drivers disabled, receiver 2 biased high

CAN transceiver, protection and optional termination









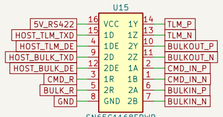




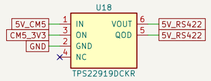




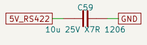
RS-422 host link



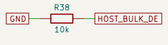






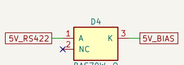












Receiver termination, fail-safe bias and 5 V to 3.3 V dividers











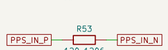












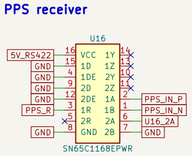








PPS receiver





Pin tables from vendor datasheets; sources in the design record.
SLANT carrier board (Raspberry Pi CM5).

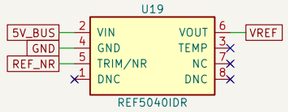
Carthage College, SLANT

Sheet: /Host interfaces: CAN and RS-422/
File: interfaces.kicad_sch

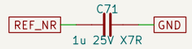
Title: Host interfaces: CAN and RS-422

Size: A2	Date: 2026-10-10	**Rev: A4-5**
KiCad E.D.A. 10.0.7		Id: 5/7

Precision 4.096 V reference (ADC VA and thermistor bias)

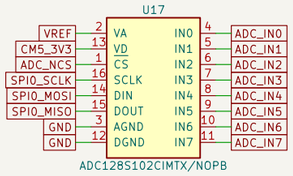




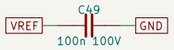




Housekeeping ADC









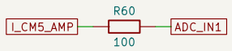


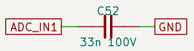
Rail voltage and current channels

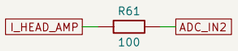






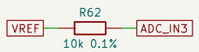






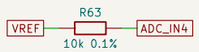


Thermistor channels



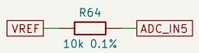








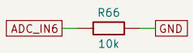




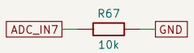














Host thermistor (read by the host, separate from payload circuits)



NOTES (value basis and items to confirm)
U19: 4.096 V, 0.05 %, 3 ppm/C, +−10 mA. VIN >= VOUT + 0.2 V = 4.30 V from 5V_BUS (always on). Load: ADC VA plus three thermistor dividers, under 4 mA. TEMP (60k source) and NC left open; DNC pins not connected.
C70: VIN bypass, 1−10 uF
C71: Noise reduction, 1 uF (datasheet layout guideline)
C48: CL: 10 uF + C72 + C49 = 11.1 uF, inside 1−50 uF; ceramic ESR is under the 1.5 Ohm maximum (series resistor optional)
U17: VA = VREF = 4.096 V is the reference: 1.00 mV per code. VD = CM5_3V3 (2.7 V to VA). SCLK 8−16 MHz.
C72: VA: 1 uF + 0.1 uF within 1 cm (ADC datasheet)
C50: VD
R58: ch 0: 5V_BUS / 2
R60: ch 1: CM5 rail current, 1.0 V/A; RC 100 Ohm / 33 nF
R61: ch 2: head rail current
R62: ch 3: thermistor bias, ratiometric to VA
RT2: under the heat block plate (remote, leads to turrets)
R63: ch 4: thermistor bias, ratiometric to VA
RT3: on the DC−DC converter +x face (remote, leads to turrets)
R64: ch 5: thermistor bias, ratiometric to VA
RT1: on the board
R65: ch 6: 5V_HEAD / 2, direct SW2 state
R67: ch 7: spare
RT4: Host thermistor (J1 23, 24): read by the host while the payload is off. Not tied to payload circuits. Not yet in the CAD.

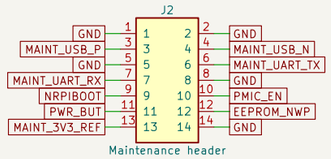
Pin tables from vendor datasheets; sources in the design record.
SLANT carrier board (Raspberry Pi CM5).
Carthage College, SLANT

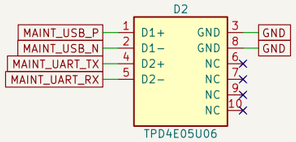
Sheet: /Housekeeping ADC, reference, rail sensing and thermistors/
File: housekeeping.kicad_sch

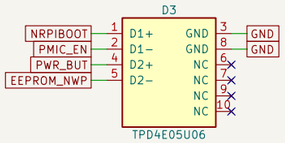
Title: Housekeeping ADC, reference, rail sensing and thermistors

Size: A3	Date: 2026−10−10	**Rev: A4−5**
KiCad E.D.A. 10.0.7		Id: 6/7

CM5 maintenance header and protection





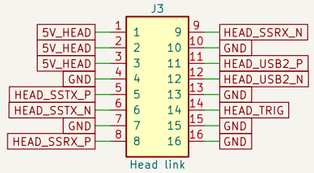






NOTES (value basis and items to confirm)
J2: Row assignment (odd/even) to confirm on drawing SD-87831-0001 sheet 8 before the footprint
D3: PMIC_EN and PWR_BUT are pulled to 5 V inside the CM5: within the 5.5 V working voltage
R68: Boot EEPROM write-protected in flight; ground equipment pulls the pin high to program
R69: Logic-level reference for the ground adapter
J3: USB 3.0 port 0 of the CM5 (TX AC-coupled on the CM5). 90 Ohm differential pairs. HEAD_TRIG (GPIO26) use TBC with XIMEA.

Flex link to the XIMEA head



Pin tables from vendor datasheets; sources in the design record.
SLANT carrier board (Raspberry Pi CM5).
Carthage College, SLANT

Sheet: /Maintenance port and head link/
File: maint_head.kicad_sch

Title: Maintenance port and head link

Size: A3	Date: 2026-10-10	**Rev: A4-5**
KiCad E.D.A. 10.0.7		Id: 7/7